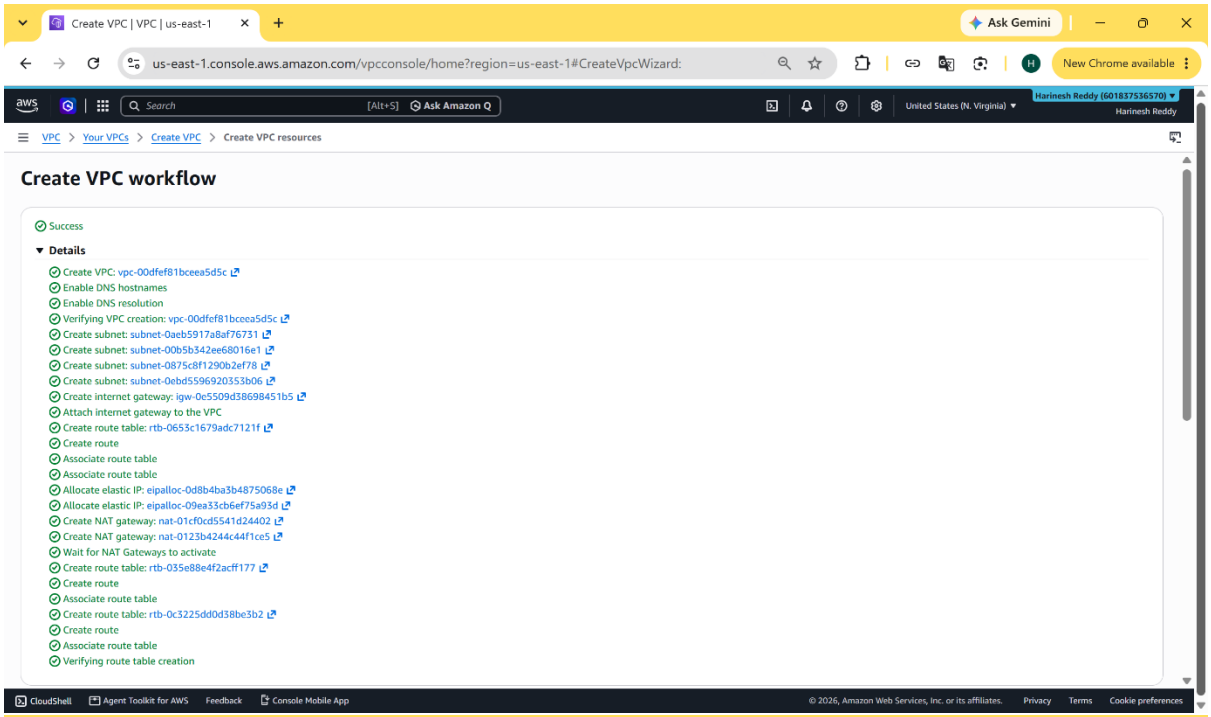
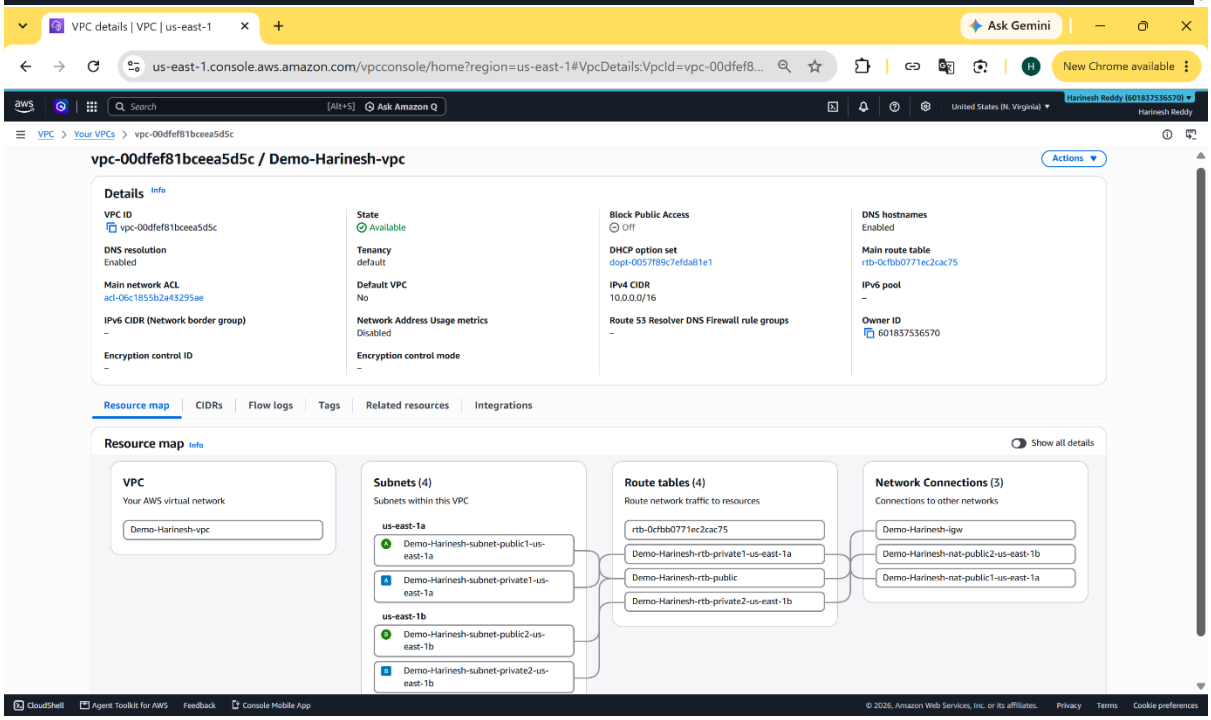
AWS Capstone Project 1

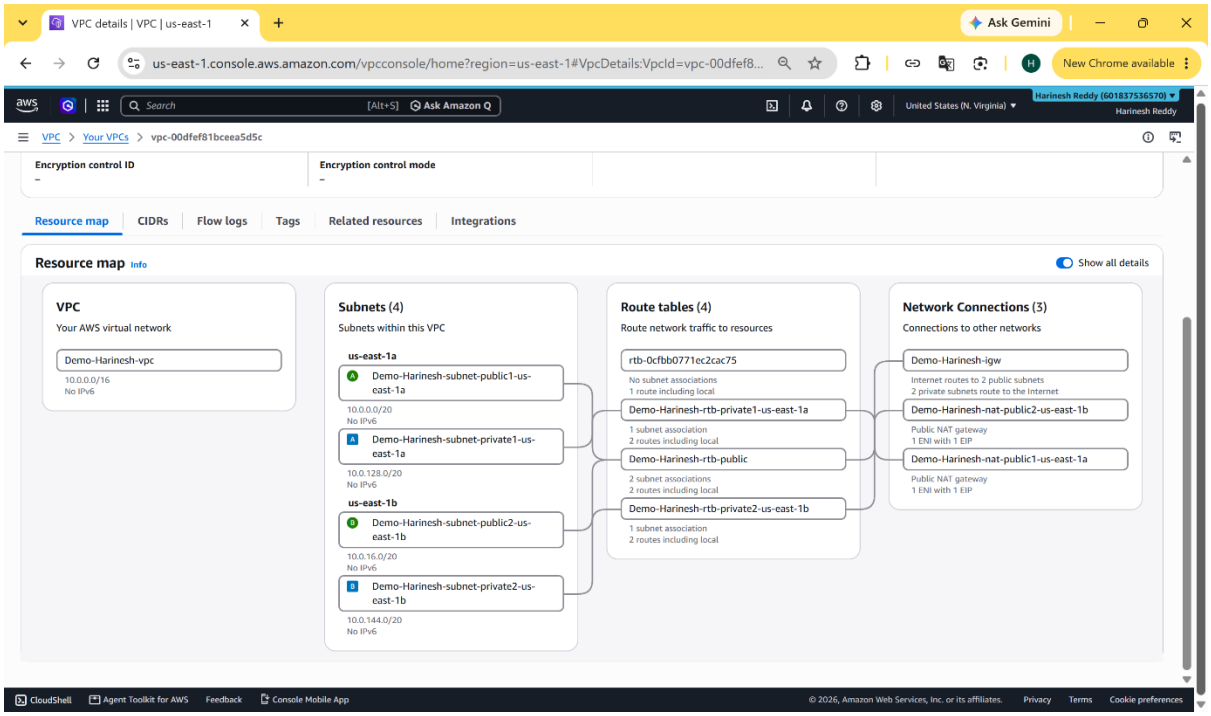
Two tier Highly Available and Highly Scalable Architecture

Solution / Screenshots

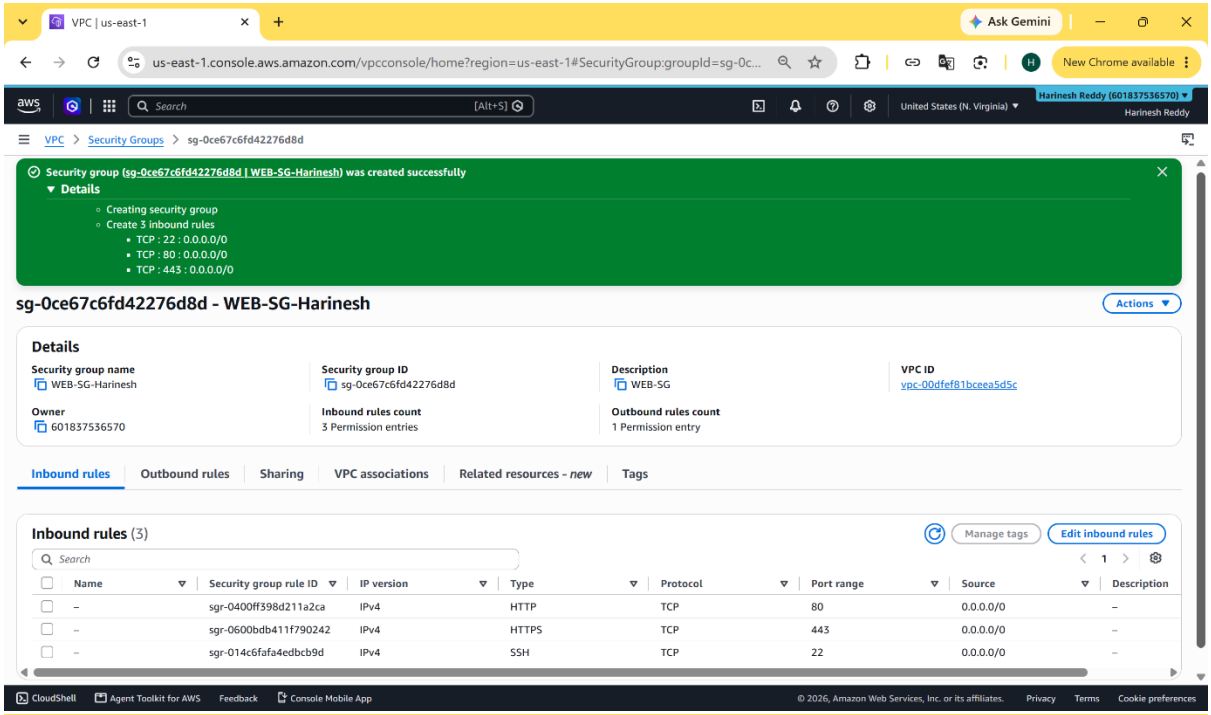
Task 1: Create a VPC:

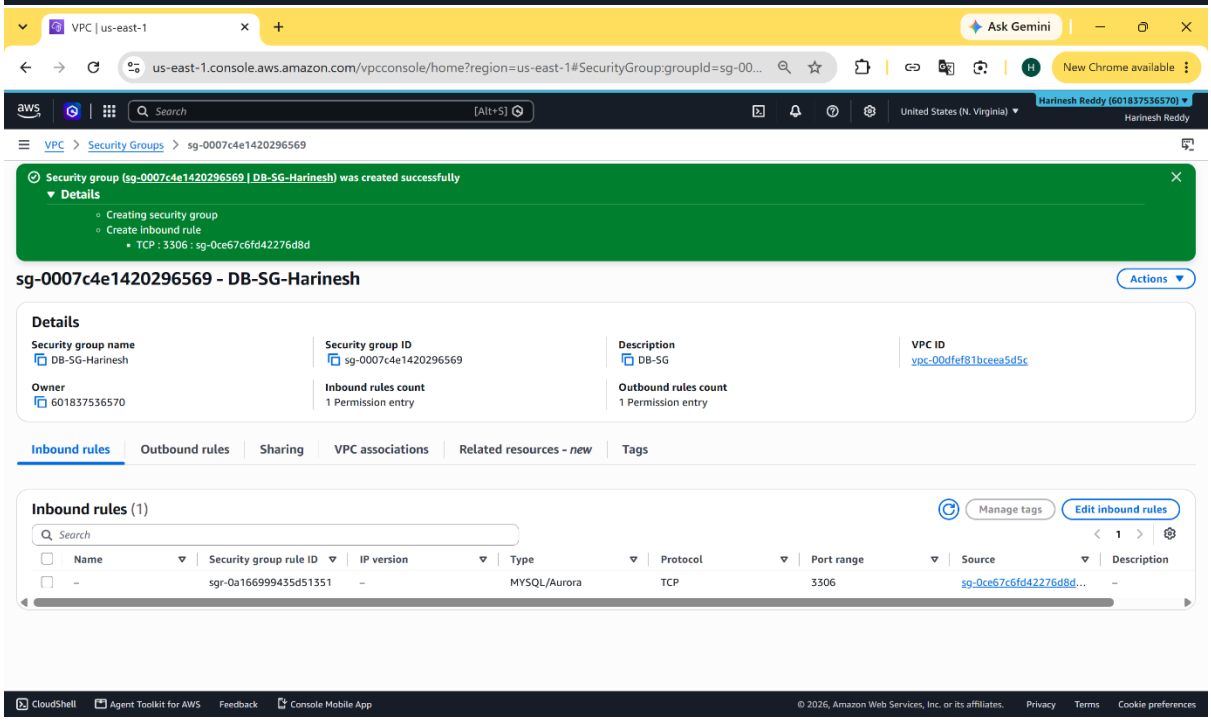




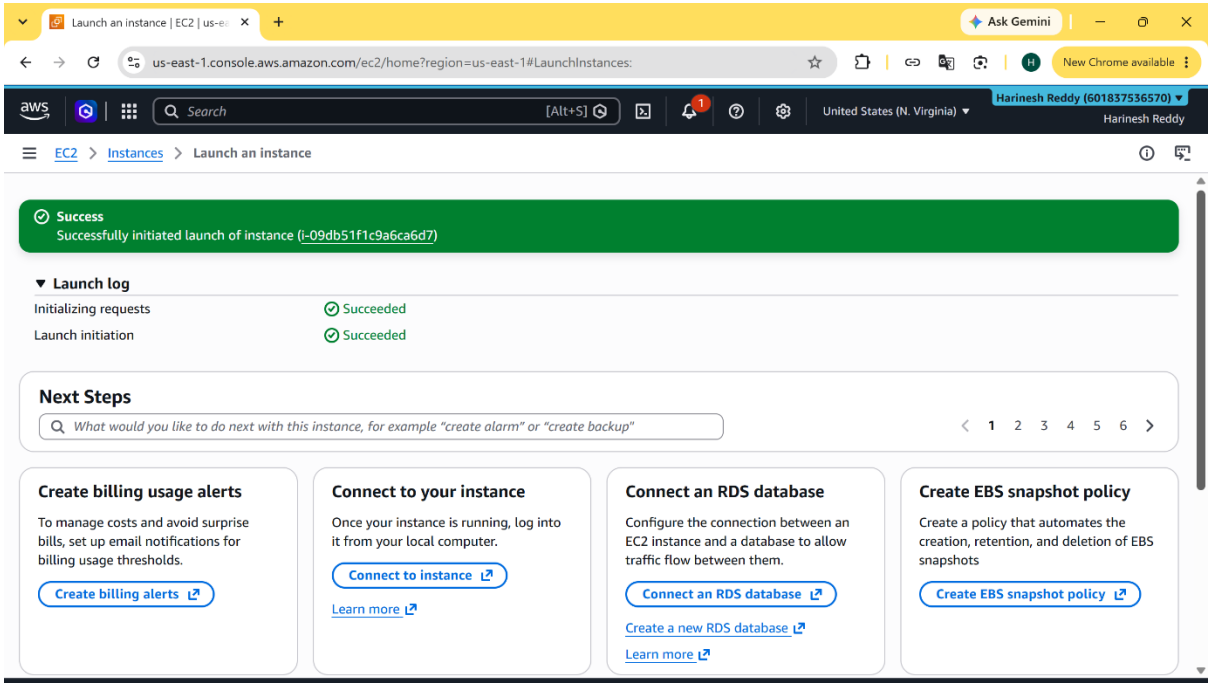

VPC details | VPC | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#VpcDetails:VpcId=vpc-00dfef8...
New Chrome available
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Harinesh Reddy (601837536570)
Harinesh Reddy
VPC > Your VPCs > vpc-00dfef81bceea5d5c
Encryption control ID
-
Encryption control mode
-
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
Resource map Info
Show all details
VPC
Your AWS virtual network
Demo-Harinesh-vpc
10.0.0.0/16
No IPv6
Subnets (4)
Subnets within this VPC
us-east-1a
Demo-Harinesh-subnet-public1-us-east-1a
10.0.0.0/20
No IPv6
Demo-Harinesh-subnet-private1-us-east-1a
10.0.128.0/20
No IPv6
us-east-1b
Demo-Harinesh-subnet-public2-us-east-1b
10.0.16.0/20
No IPv6
Demo-Harinesh-subnet-private2-us-east-1b
10.0.144.0/20
No IPv6
Route tables (4)
Route network traffic to resources
rtb-0cfbb0771ec2cac75
No subnet associations
1 route including local
Demo-Harinesh-rtb-private1-us-east-1a
1 subnet association
2 routes including local
Demo-Harinesh-rtb-public
2 subnet associations
2 routes including local
Demo-Harinesh-rtb-private2-us-east-1b
1 subnet association
2 routes including local
Network Connections (3)
Connections to other networks
Demo-Harinesh-igw
Internet routes to 2 public subnets
2 private subnets route to the Internet
Demo-Harinesh-nat-public2-us-east-1b
Public NAT gateway
1 ENI with 1 EIP
Demo-Harinesh-nat-public1-us-east-1a
Public NAT gateway
1 ENI with 1 EIP
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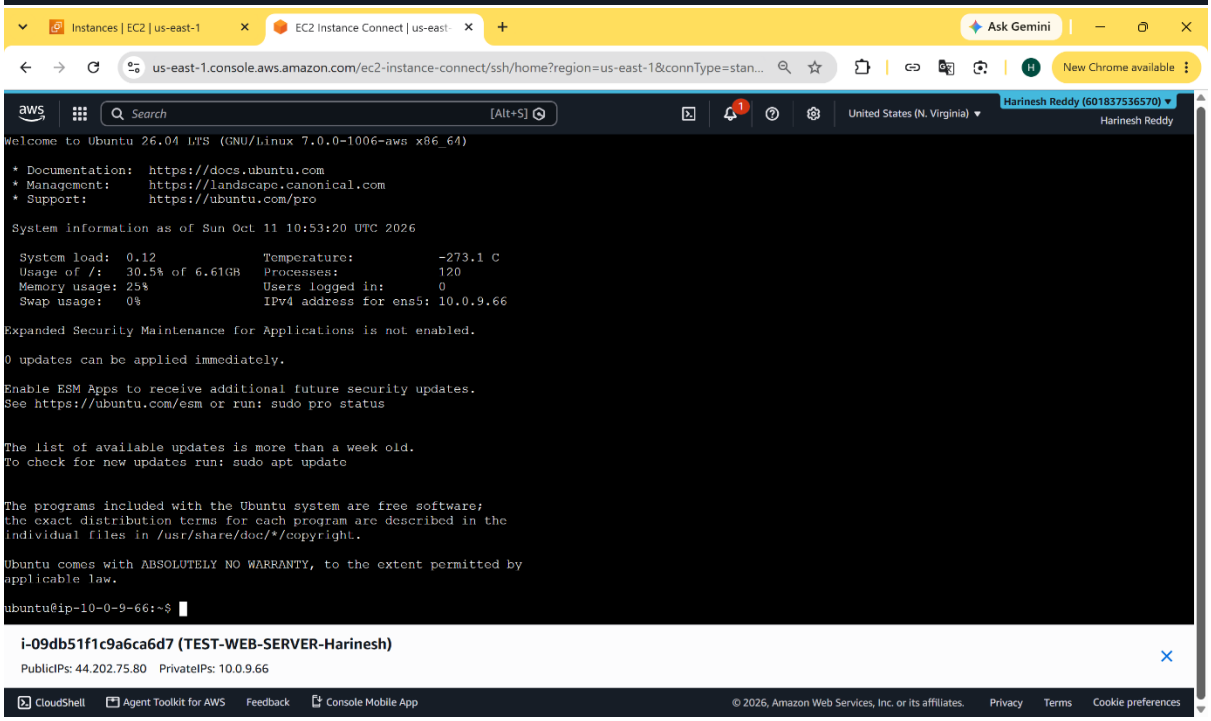
Task 2: Create 2 Security Groups:





Task 3: Create, connect, and configure Instance:

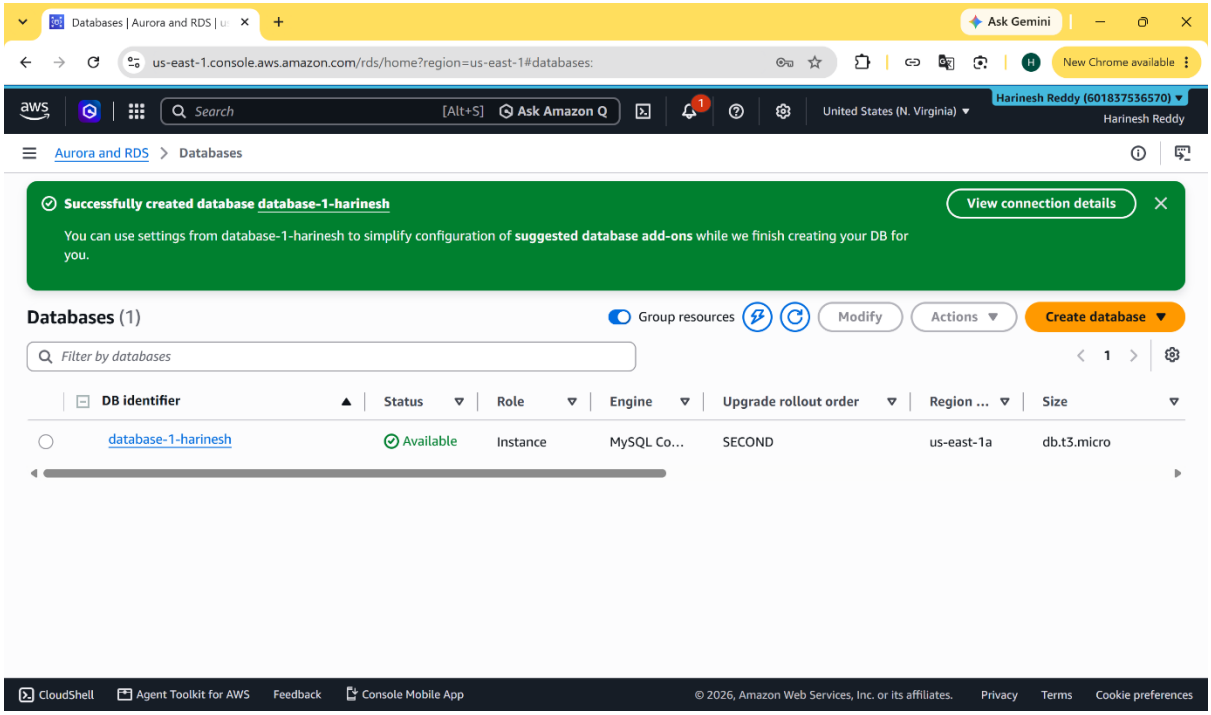


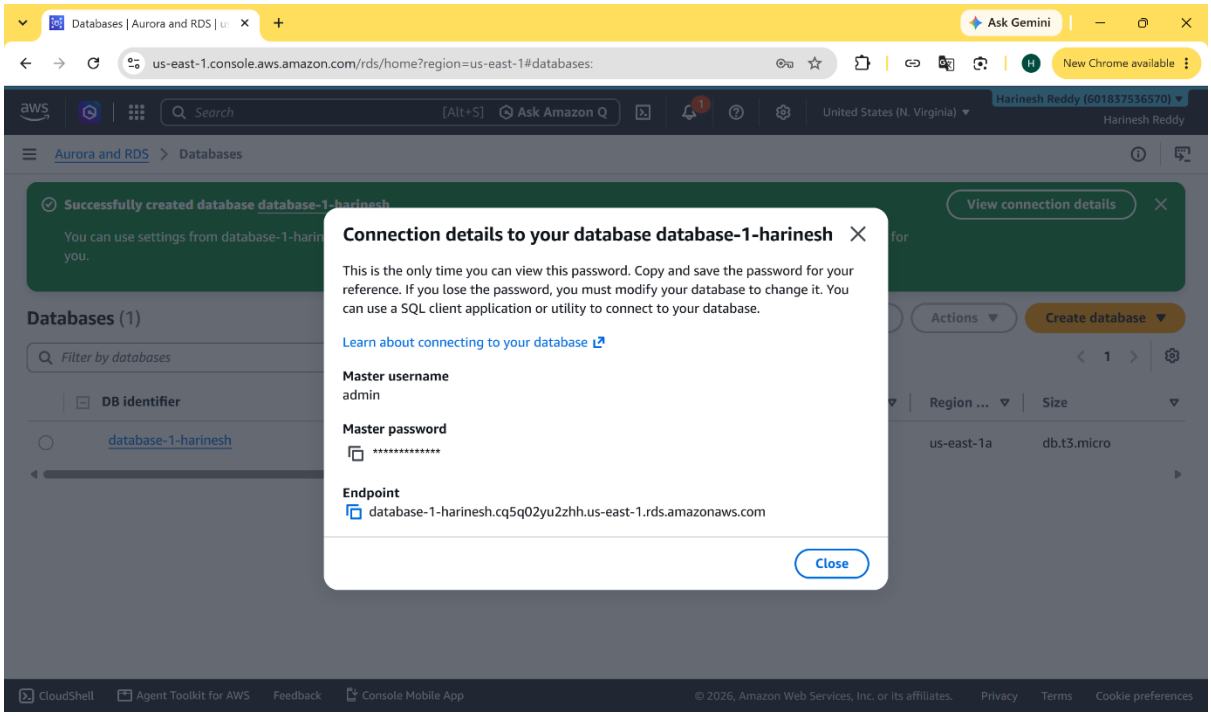




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Task 4: Create, RDS MySQL database:

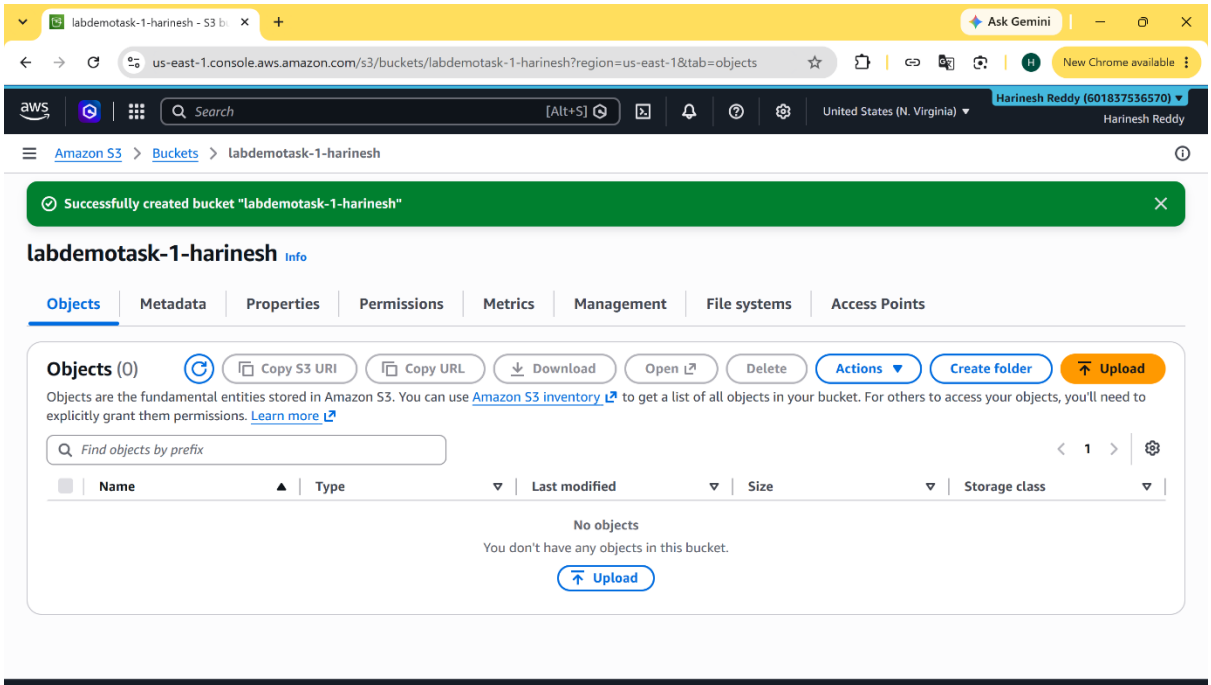


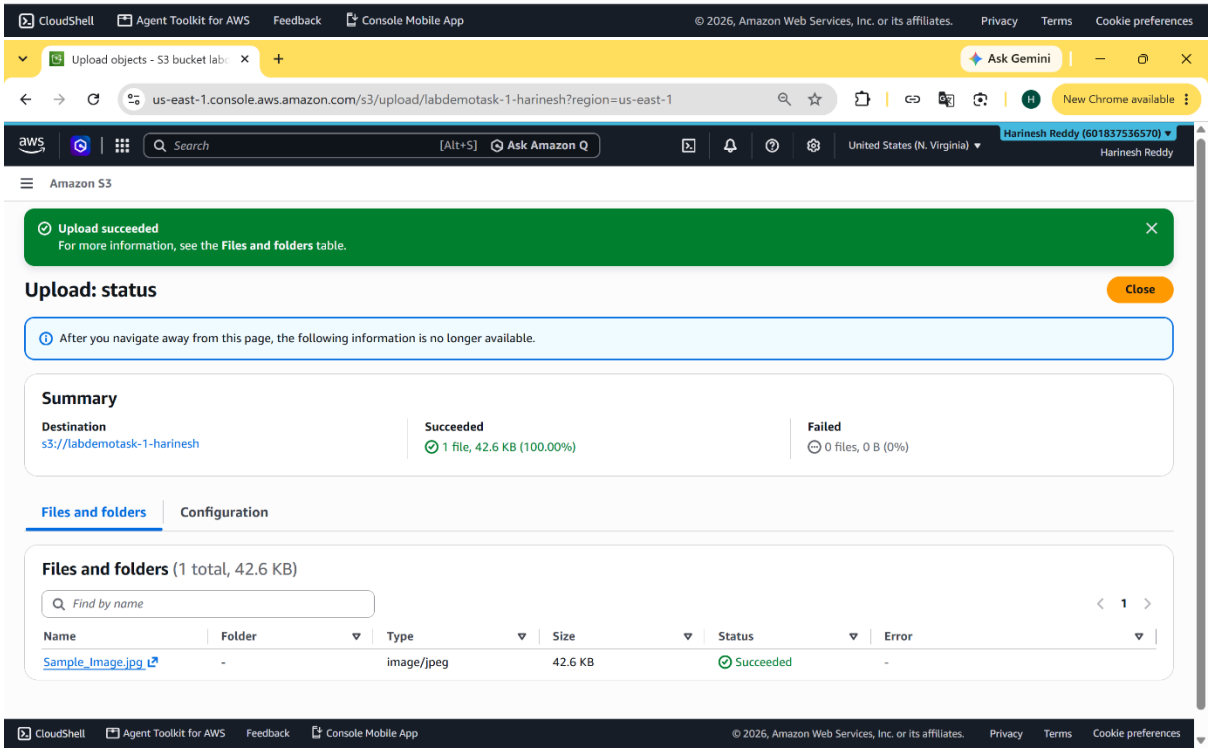


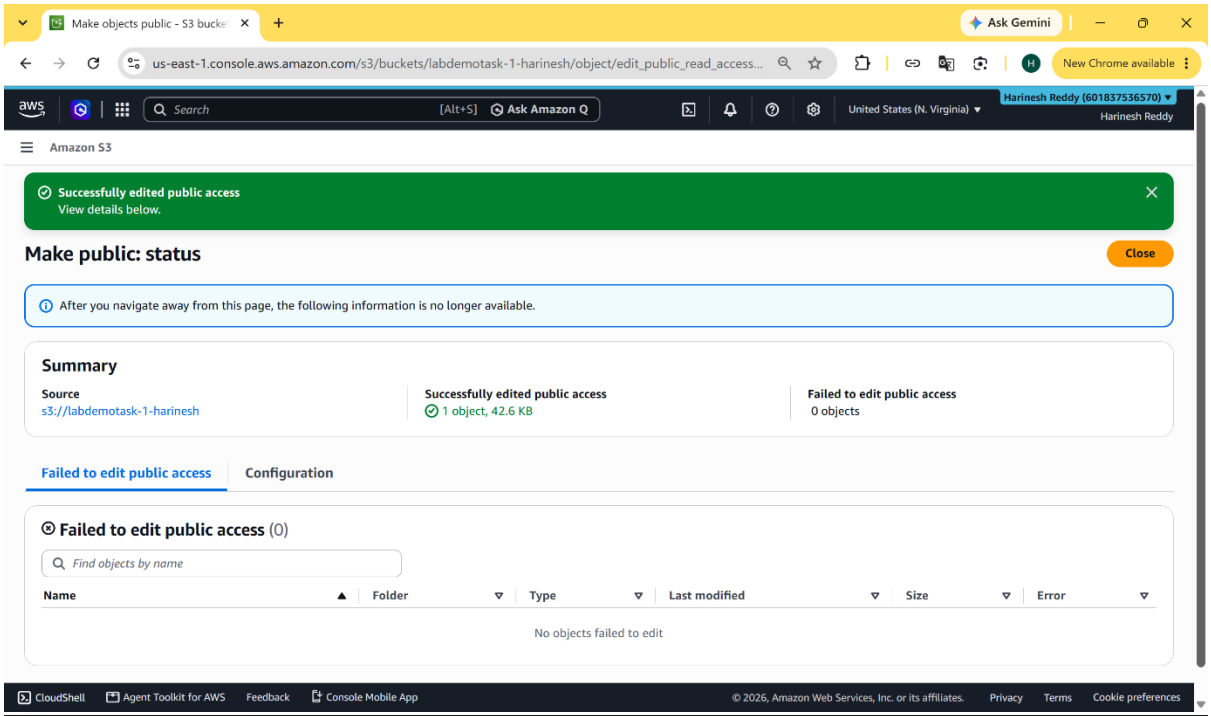
Task 5: Configure Instance to connect to MySQL Database:

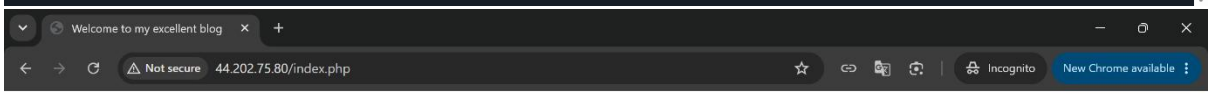


Task 6: Configure Instance to get the images from Amazon S3 on website:







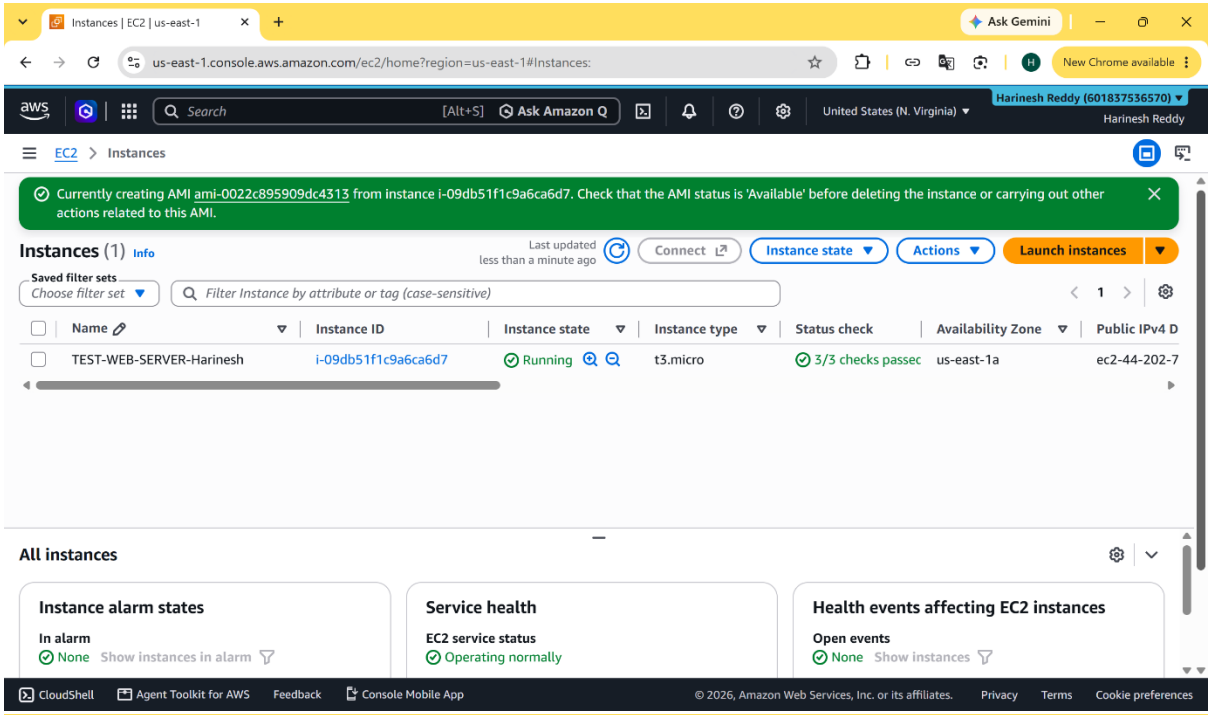


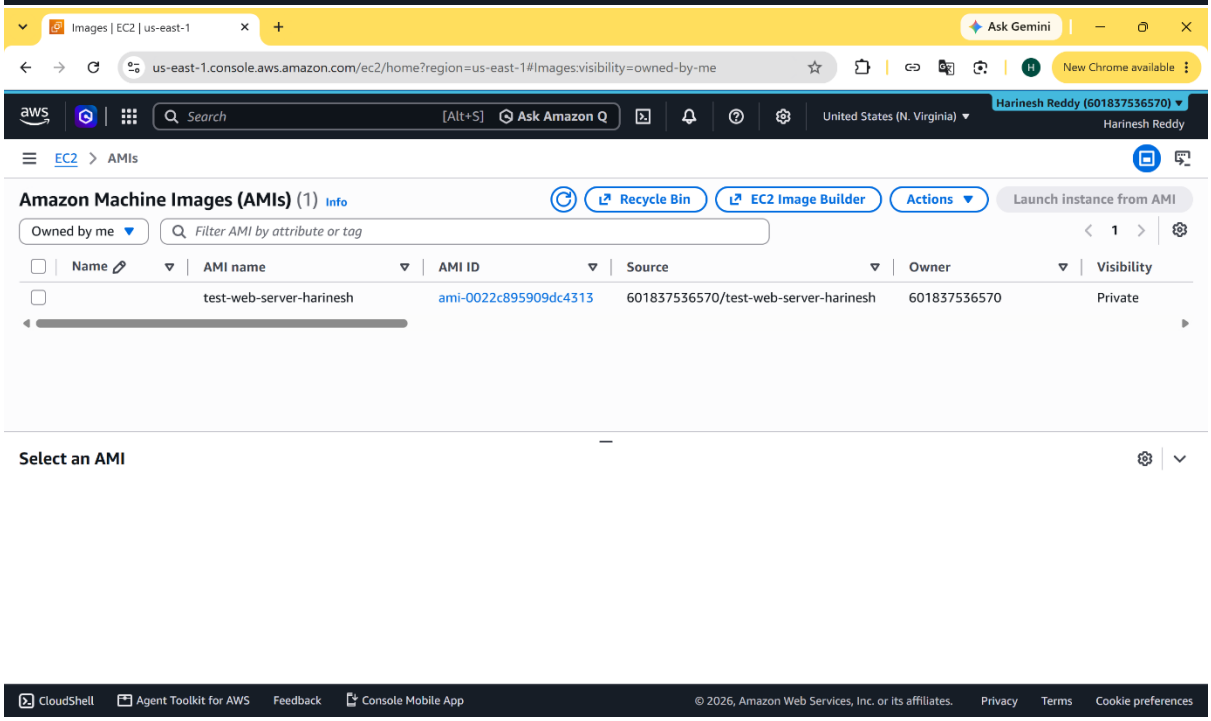
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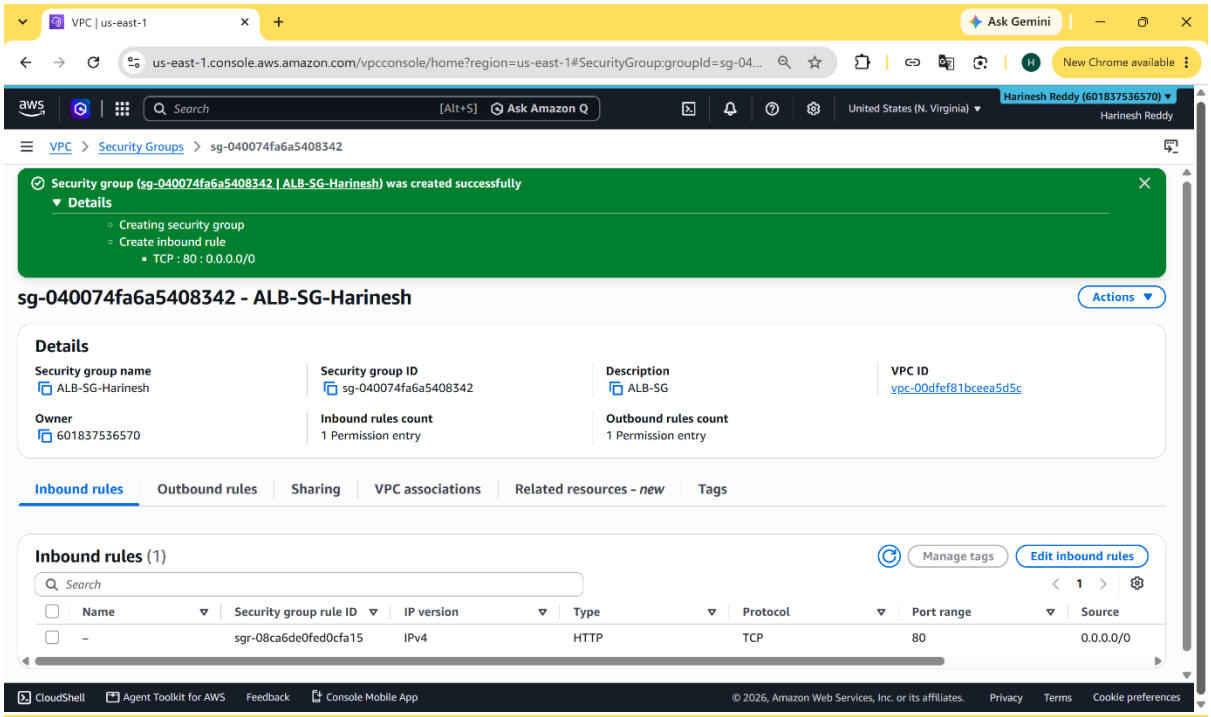
Database connection succeeded.

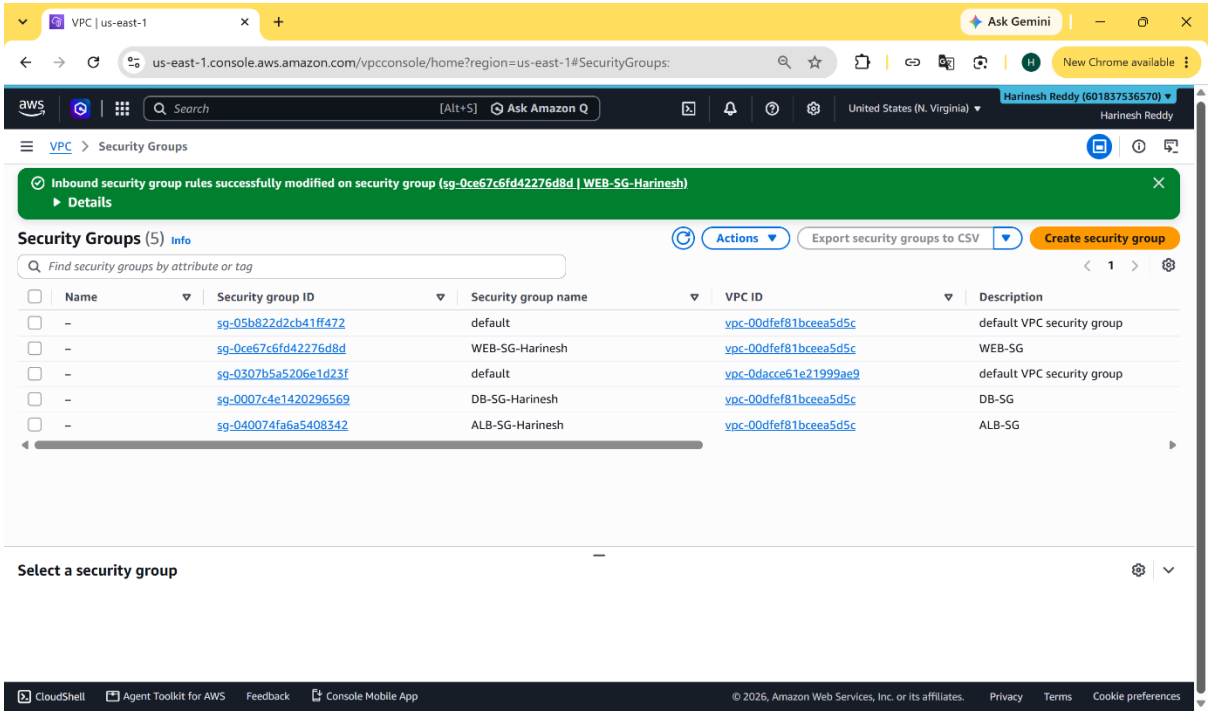
Task 7: Create AMI image of Instance:



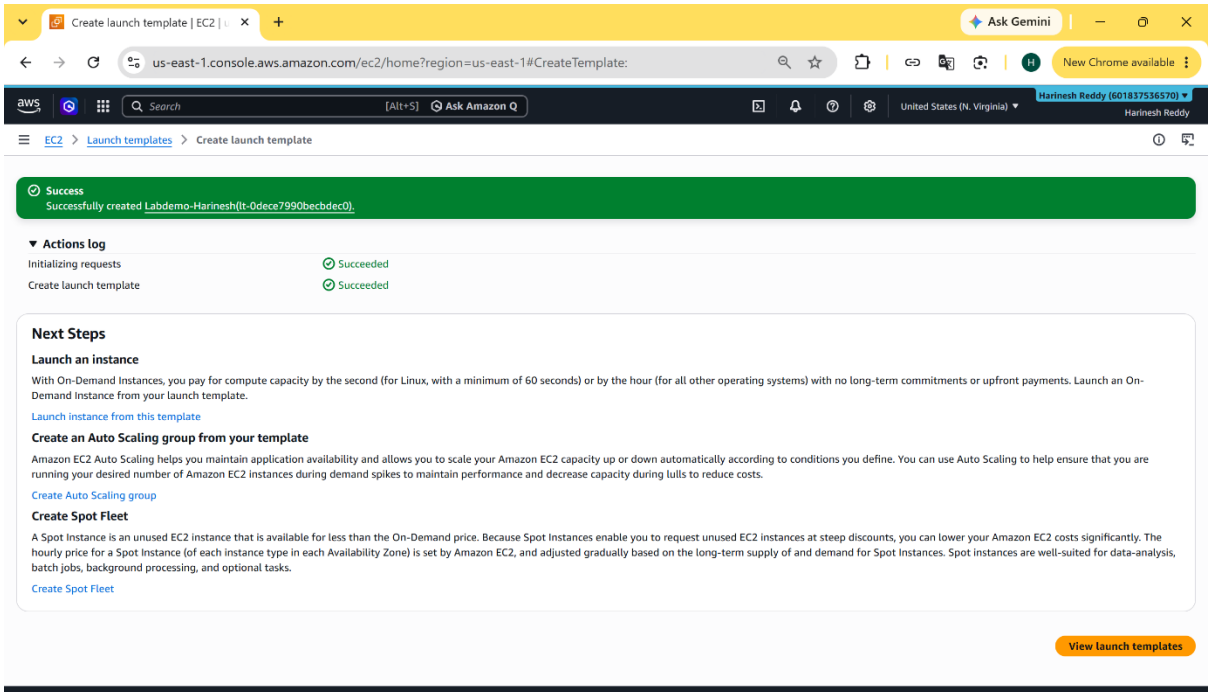


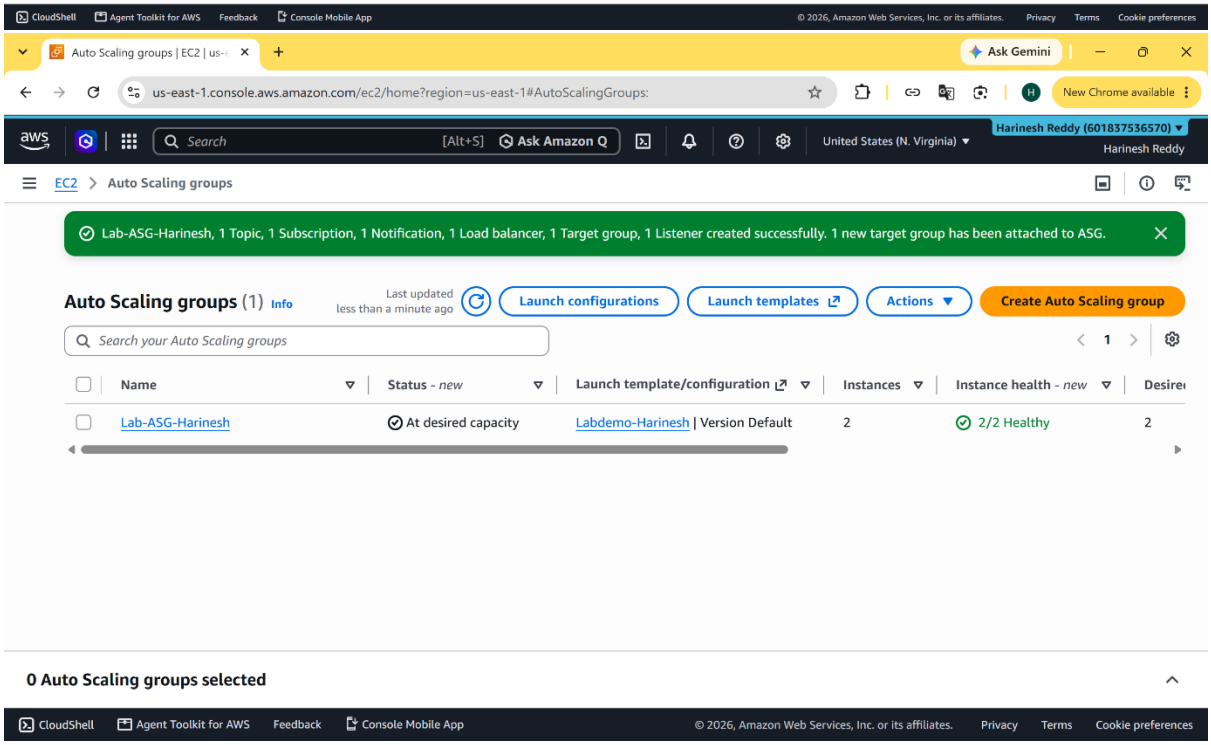
Task 8: Create security group for ALB and modify security group of instance:

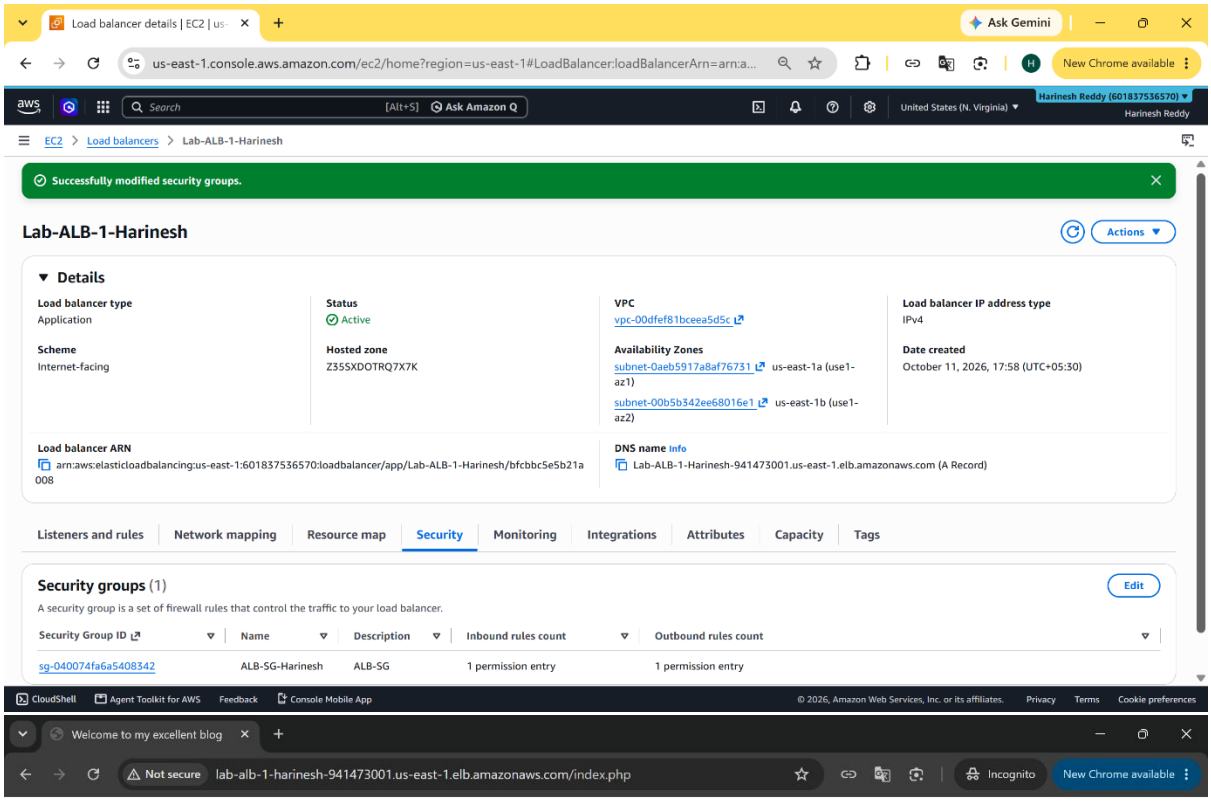




Task 9: Implement high availability and scalability to the application:





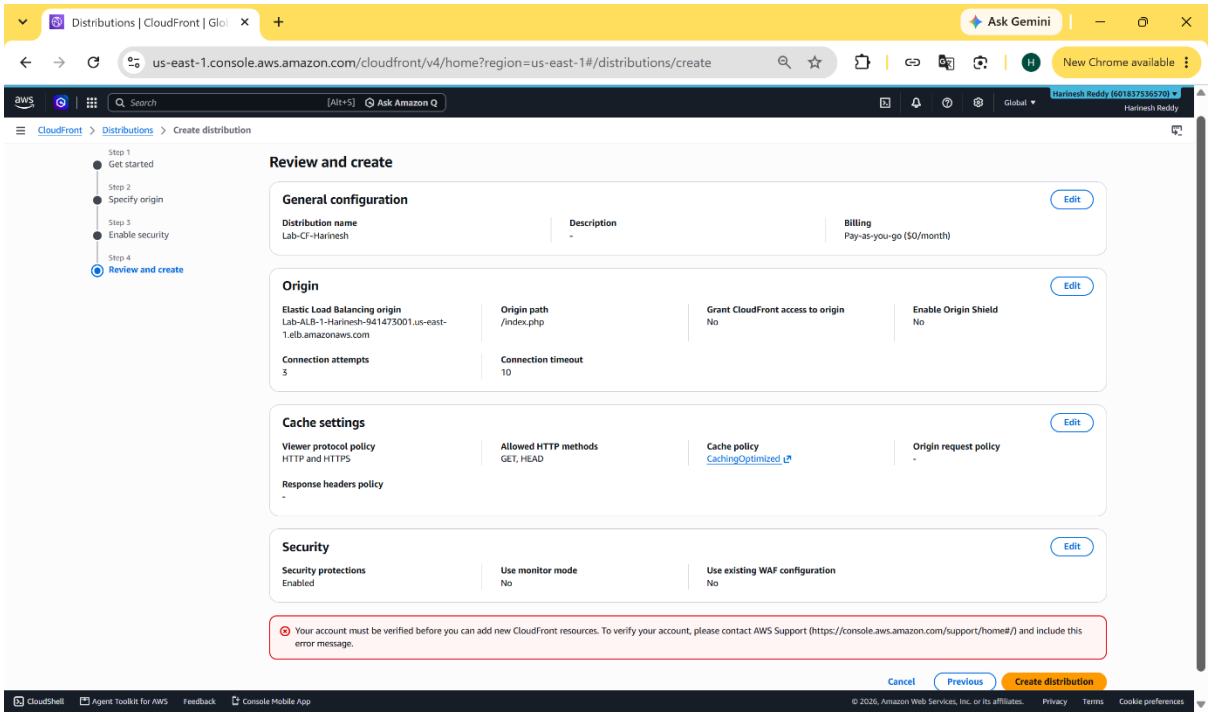


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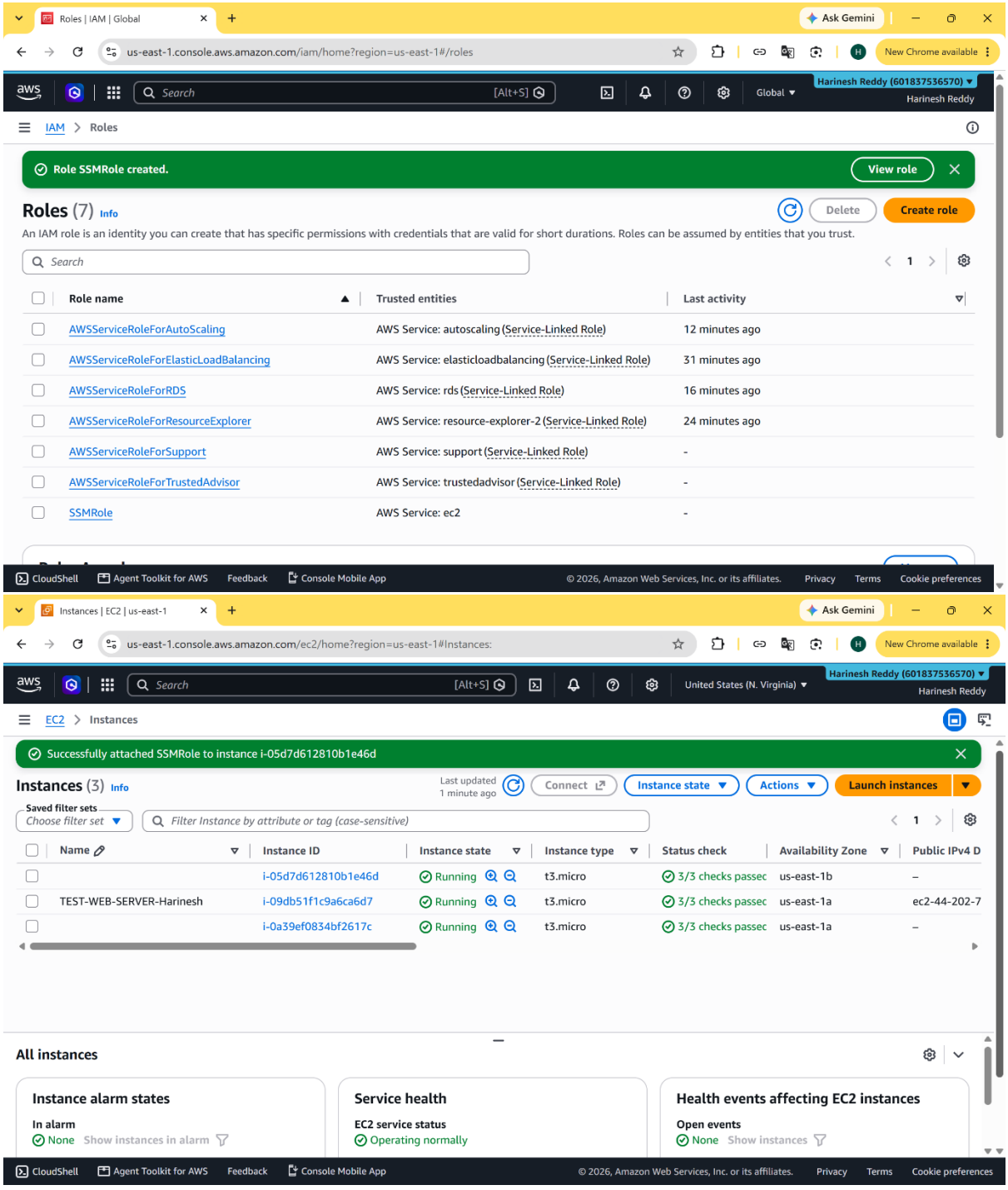


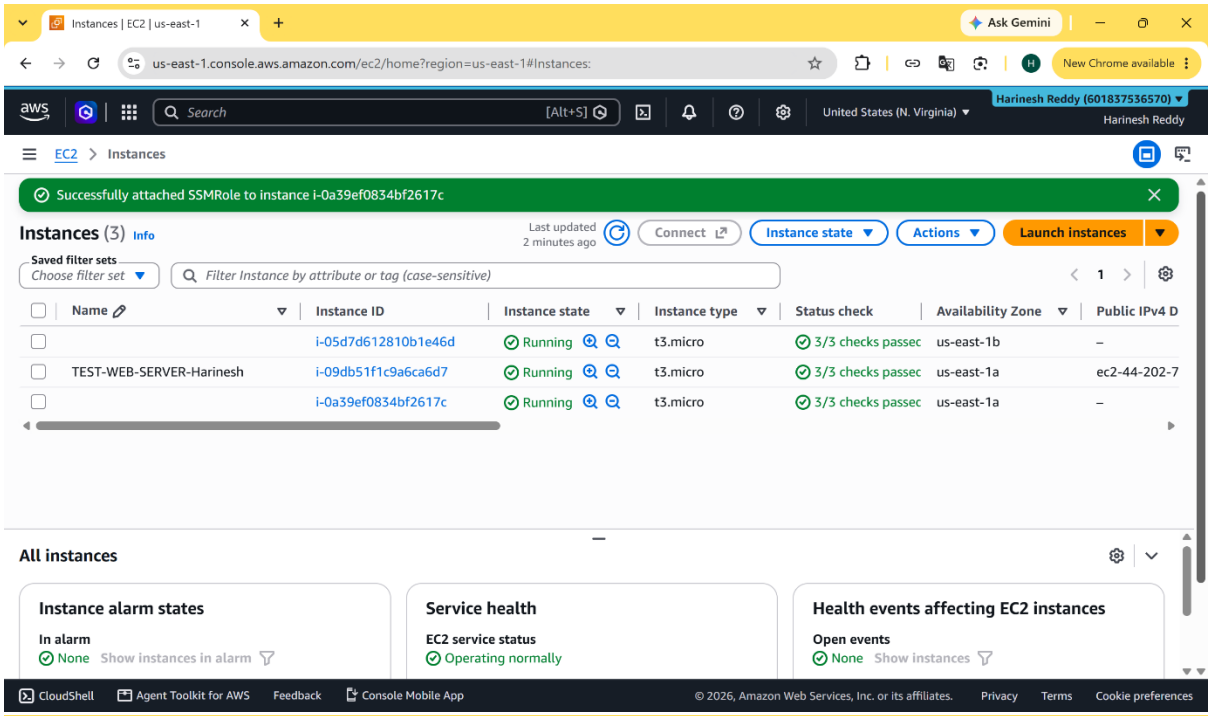
Database connection succeeded.

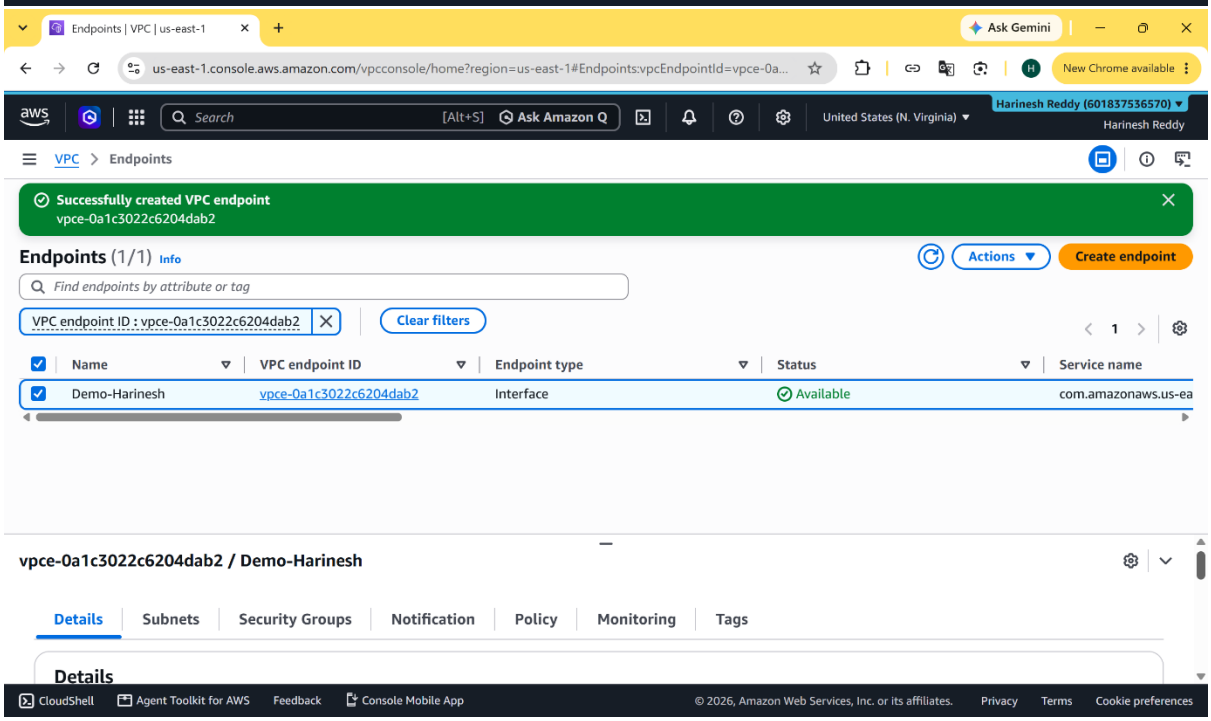
Task 10: Implement low latency and security from DDoS attack to the application:

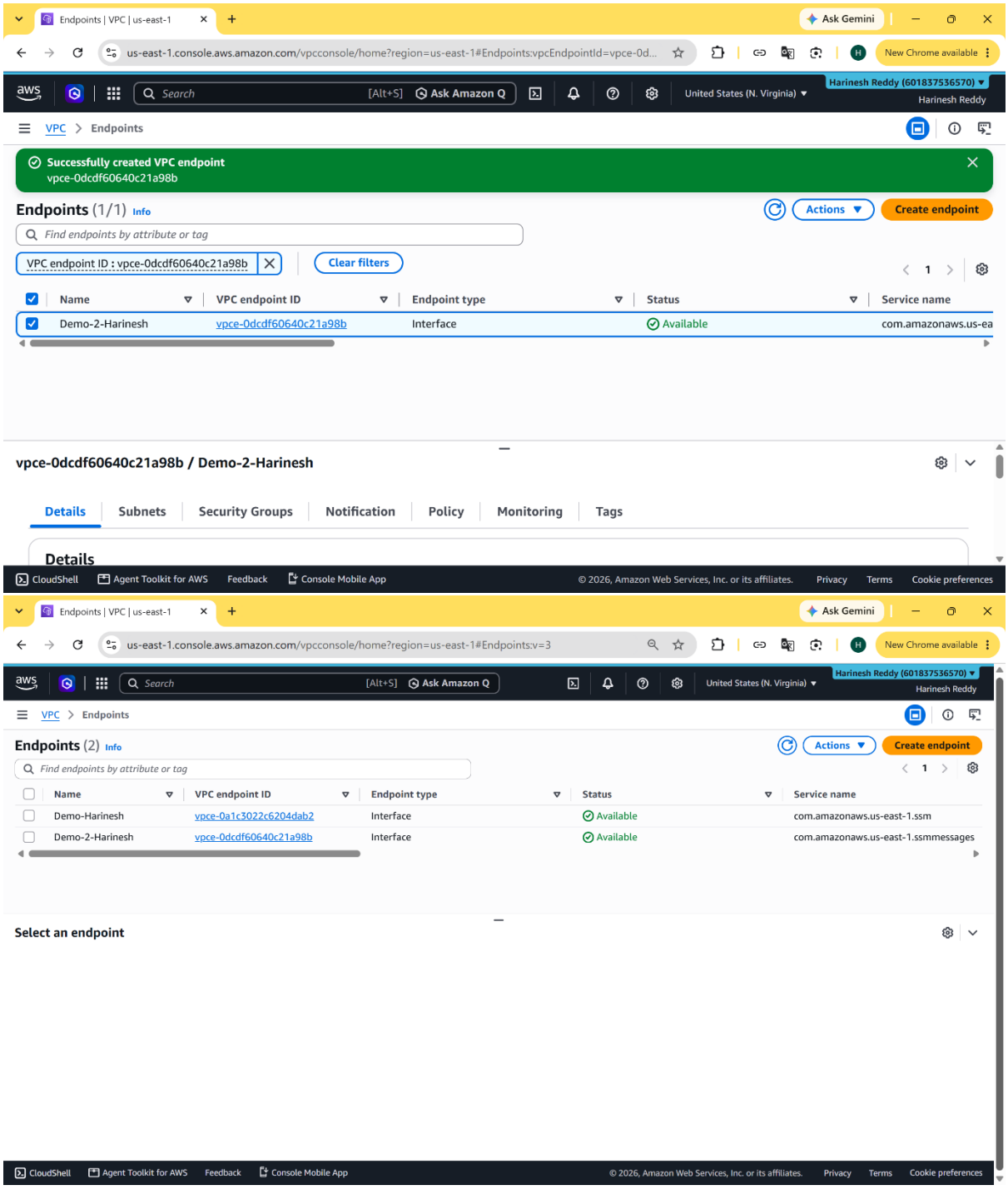


Task 11: Implement secure patching for EC2 instances:

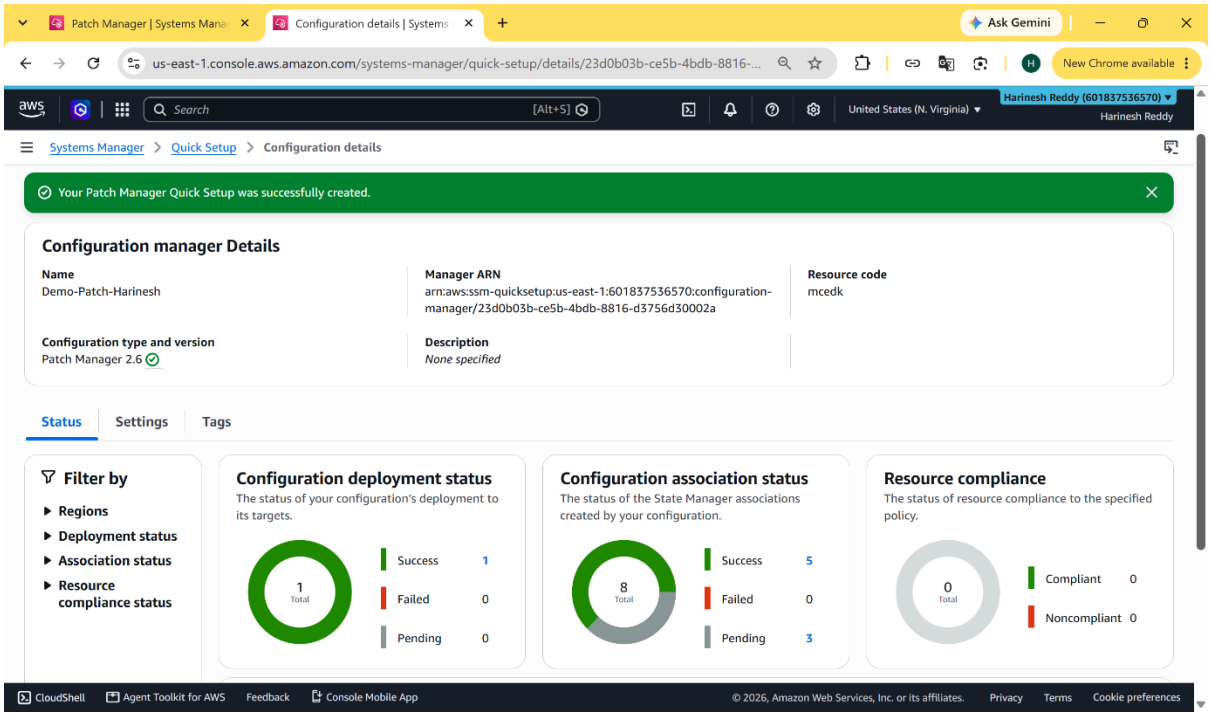



Instances | EC2 | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Harinesh Reddy (601837536570)
Harinesh Reddy
EC2 > Instances
Successfully attached SSMRole to instance i-0a39ef0834bf2617c
Instances (3) Info
Last updated 2 minutes ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 D
i-05d7d612810b1e46d Running t3.micro 3/3 checks passec us-east-1b –
TEST-WEB-SERVER-Harinesh i-09db51f1c9a6ca6d7 Running t3.micro 3/3 checks passec us-east-1a ec2-44-202-7
i-0a39ef0834bf2617c Running t3.micro 3/3 checks passec us-east-1a –
All instances
Instance alarm states
In alarm
None Show instances in alarm
Service health
EC2 service status
Operating normally
Health events affecting EC2 instances
Open events
None Show instances
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Privacy
Terms
Cookie preferences


Endpoints | VPC | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#Endpoints:vpcEndpointId=vpce-0a...
New Chrome available
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Harinesh Reddy (601837536570)
Harinesh Reddy
VPC > Endpoints
Successfully created VPC endpoint
vpce-0a1c3022c6204dab2
Endpoints (1/1) Info
Actions
Create endpoint
Find endpoints by attribute or tag
VPC endpoint ID : vpce-0a1c3022c6204dab2
Clear filters
Name
VPC endpoint ID
Endpoint type
Status
Service name
Demo-Harinesh vpce-0a1c3022c6204dab2 Interface Available com.amazonaws.us-ea
vpce-0a1c3022c6204dab2 / Demo-Harinesh
Details
Subnets
Security Groups
Notification
Policy
Monitoring
Tags
Details
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Cookie preferences


Endpoints | VPC | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#Endpoints:vpcEndpointId=vpce-0d...
New Chrome available
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Harinesh Reddy (601837536570)
Harinesh Reddy
VPC > Endpoints
Successfully created VPC endpoint
vpce-0dcdf60640c21a98b
Endpoints (1/1) Info
Actions
Create endpoint
Find endpoints by attribute or tag
VPC endpoint ID : vpce-0dcdf60640c21a98b
Clear filters
Name
VPC endpoint ID
Endpoint type
Status
Service name
Demo-2-Harinesh
vpce-0dcdf60640c21a98b
Interface
Available
com.amazonaws.us-ea
vpce-0dcdf60640c21a98b / Demo-2-Harinesh
Details
Subnets
Security Groups
Notification
Policy
Monitoring
Tags
Details
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Privacy
Terms
Cookie preferences

Endpoints | VPC | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#Endpoints:v=3
New Chrome available
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Harinesh Reddy (601837536570)
Harinesh Reddy
VPC > Endpoints
Endpoints (2) Info
Actions
Create endpoint
Find endpoints by attribute or tag
Name
VPC endpoint ID
Endpoint type
Status
Service name
Demo-Harinesh
vpce-0a1c3022c6204dab2
Interface
Available
com.amazonaws.us-east-1.ssm
Demo-2-Harinesh
vpce-0dcdf60640c21a98b
Interface
Available
com.amazonaws.us-east-1.ssmmessages
Select an endpoint
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Privacy
Terms
Cookie preferences


Patch Manager | Systems Mana
Configuration details | Systems
Ask Gemini
us-east-1.console.aws.amazon.com/systems-manager/quick-setup/details/23d0b03b-ce5b-4bdb-8816-...
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Harinesh Reddy (601837536570)
Harinesh Reddy
Systems Manager
Quick Setup
Configuration details
Your Patch Manager Quick Setup was successfully created.
Configuration manager Details
Name
Demo-Patch-Harinesh
Manager ARN
arn:aws:ssm-quicksetup:us-east-1:601837536570:configuration-manager/23d0b03b-ce5b-4bdb-8816-d3756d30002a
Resource code
mcedk
Configuration type and version
Patch Manager 2.6
Description
None specified
Status
Settings
Tags
Filter by
Regions
Deployment status
Association status
Resource compliance status
Configuration deployment status
The status of your configuration's deployment to its targets.
1
Total
Success 1
Failed 0
Pending 0
Configuration association status
The status of the State Manager associations created by your configuration.
8
Total
Success 5
Failed 0
Pending 3
Resource compliance
The status of resource compliance to the specified policy.
0
Total
Compliant 0
Noncompliant 0
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Task 12: Monitor CPU Utilization of webapp servers:

